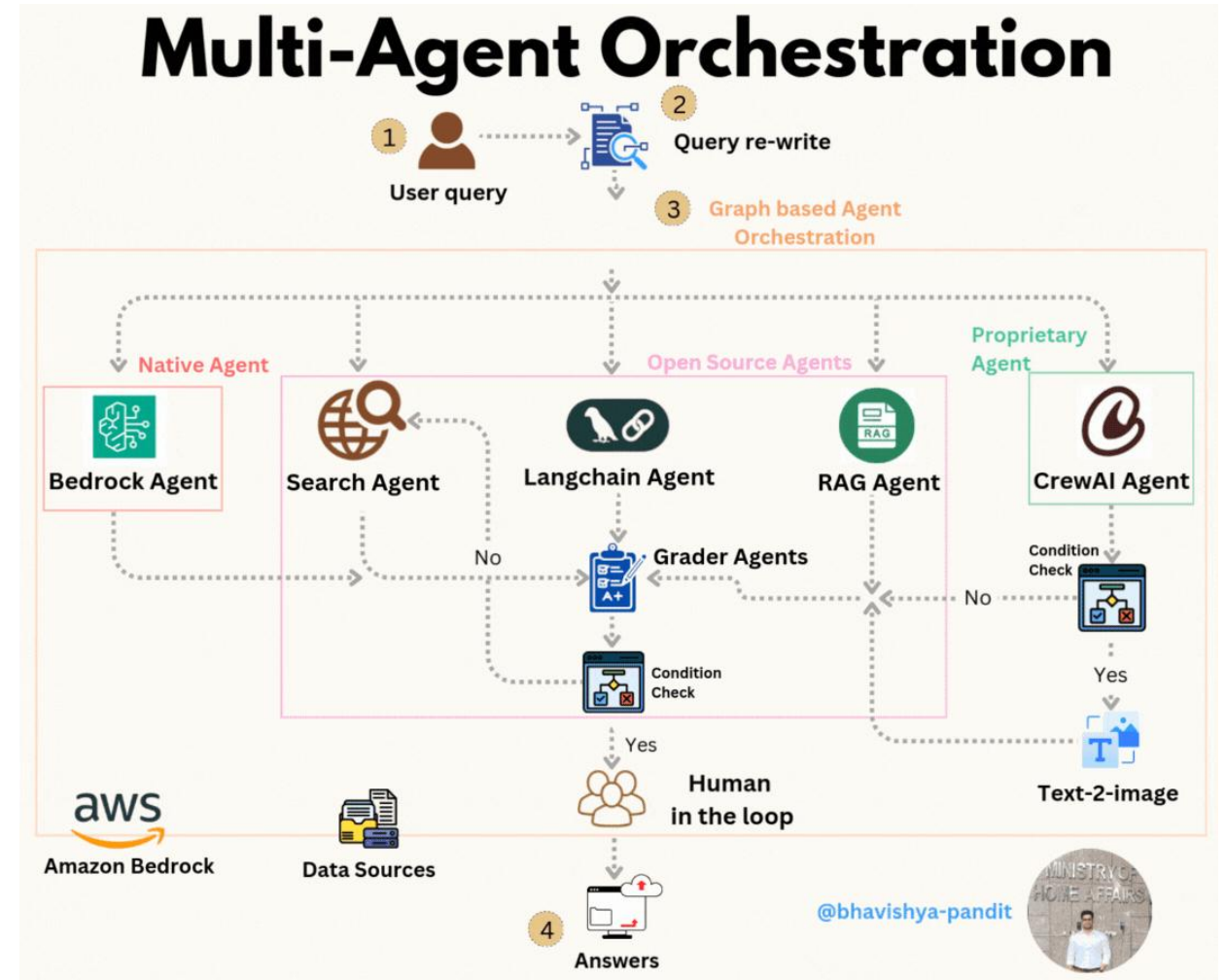
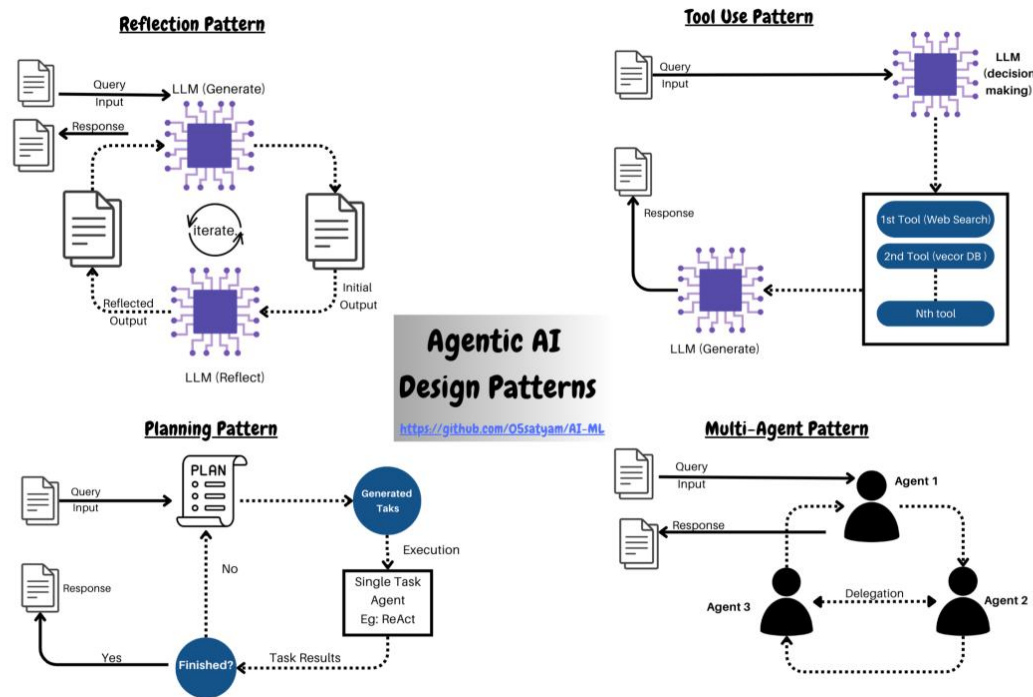


MODULE 1: MULTI-AGENT ARCHITECTURES (THE ORCHESTRATOR)

- **Goal:** Mastering the orchestration of multiple specialized AI agents to handle complex business requests.
- **Core Philosophy:** Move from single-model prompting to **Role-Based Agent Design**, where specific personas have unique constraints and domain knowledge.



ARCHITECTURAL PATTERNS



- **Autonomous Task Decomposition:** Engineering "Manager" agents that break a complex goal into a multi-step execution plan.
- **Handoff Protocols:** Establishing how agents transfer structured data (JSON) and maintain "Shared Memory" across long workflows.
- **Flow Types:** Comparing **Hierarchical** systems (agents reporting to a supervisor) vs. **Collaborative** systems (agents negotiating as peers).

THE REFLEXION PATTERN

- **Self-Correction Loops:** Building loops where a second agent reviews and critiques the first agent's output before finalization.
- **Quality Auditing:** Using specialized "Quality Auditor" personas to enforce standards and reduce hallucinations.
- **Task 1 Deliverable:** A functional autonomous team consisting of a Researcher, Writer, and Manager.

